Watching Language Steer a Quantum Optimizer

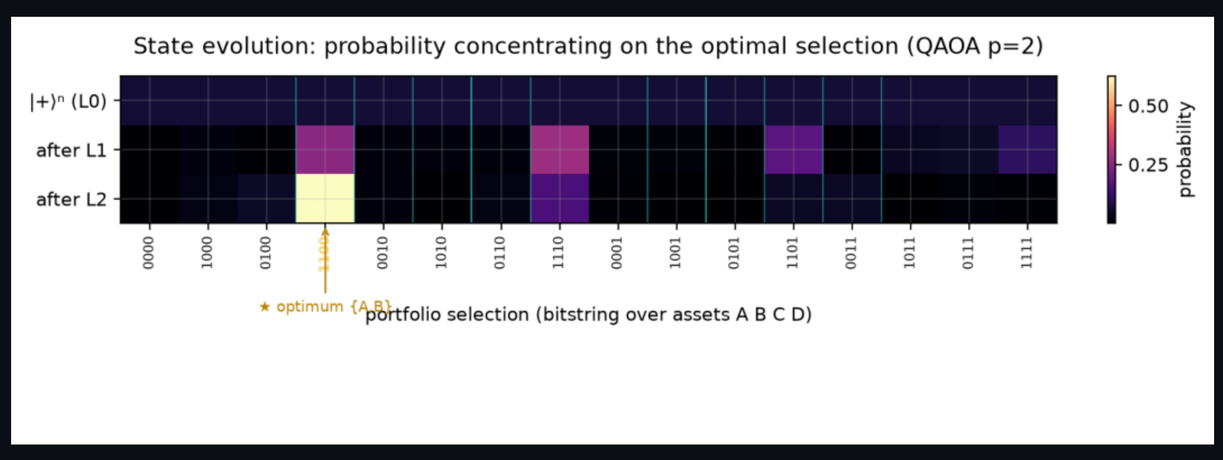
State-Evolution Visualization of LLM-Conditioned QAOA for Cardinality-Constrained Portfolio Selection

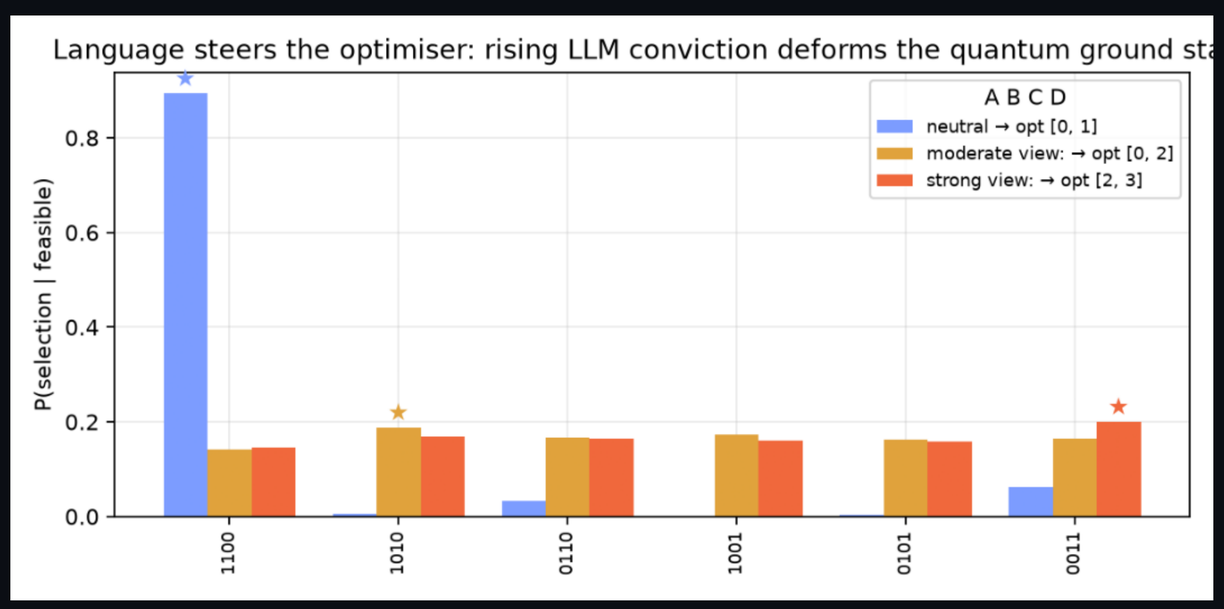
QNLPAI 2026 · Quantum Computing for AI — Quantum Optimization (QNLP bridge) · QLPC Project

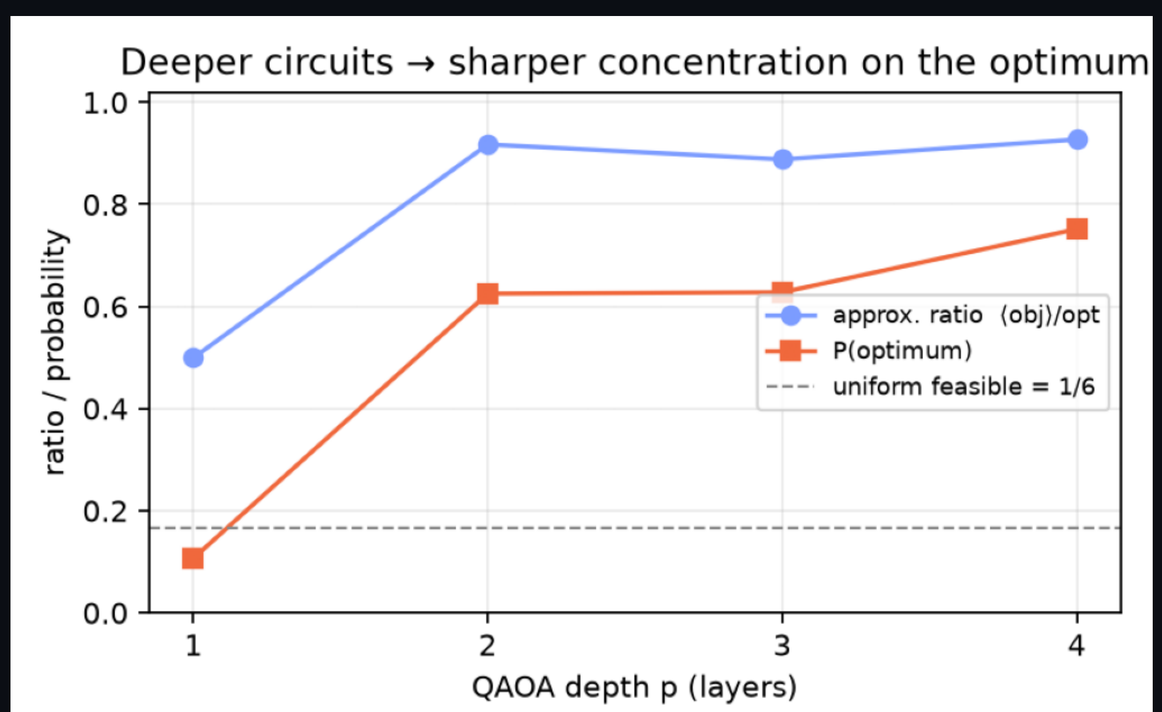
PROBLEM QAOA is reported as a black box — one bitstring, one ratio — hiding HOW the state decides, with no port for the language-shaped views that drive real allocation.

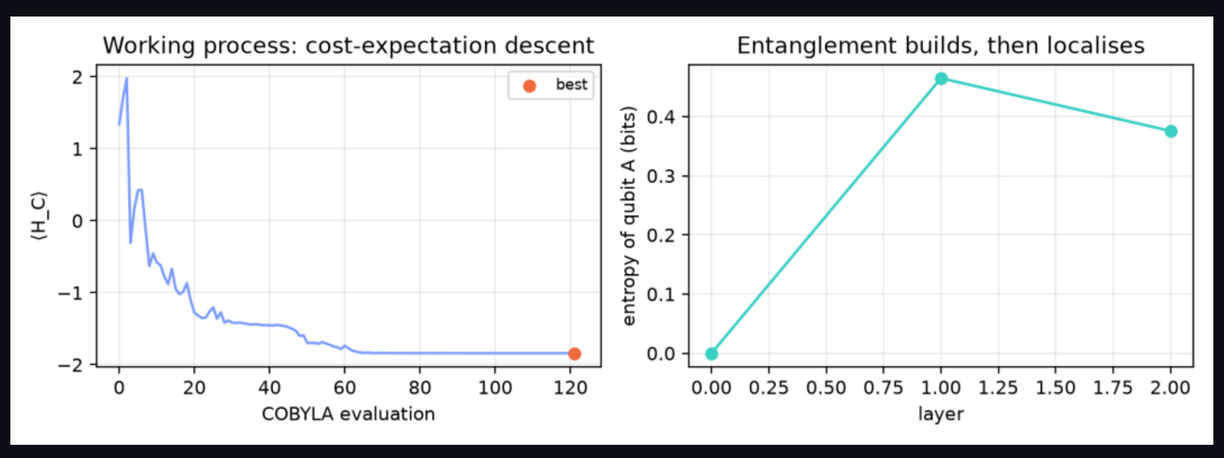
GAP LLM views exist for CLASSICAL Black-Litterman (Lee+ 2025); quantum views exist but are QML-predicted numbers solved by VQE (Chen+ 2023) — no language, no view of the quantum state.

TWO OBJECTIVES O1 encode an LLM natural-language view as a tilt on expected returns that deforms the QAOA cost Hamiltonian; O2 snapshot the statevector each layer to make the language-steering auditable. (Exact statevector simulation; IBM Qiskit.)









RESULTS (exact statevector, fixed seeds)

O2 State concentrates on the optimum: P(opt) 0.062 → 0.63 at p=2 = 10.0× a uniform guess; approx-ratio 0.92; exact optimum recovered.

O1 Language steers the ground state: optimum {A,B} → {A,C} → {C,D} as LLM conviction rises; the QAOA modal selection tracks it in all 3 regimes.

HONEST LIMITS small (n=4), well-separated, exactly-simulated instances; no scaling / hardware / quantum-advantage claim (brute force is the oracle); additive language tilt.

Reproducible: python -m paper.experiments (Qiskit 2.x + Aer) · circuits.py · experiments.py · results.json